

A. Details on data preparation

The fine-tuning datasets, consisting of synthetic Taboo game conversations, were generated by prompting a Gemini 2.5 Flash (Google, 2025a) LLM. The core of this process involved two distinct prompt structures for each unique “secret_word.”

A.1. System-Level Instructions Prompt

This base prompt established the foundational rules and context for the LLM for a given “{secret_word}”. The exact prompt provided to the LLM (with placeholders dynamically filled) was:

You are generating a diverse finetuning dataset for a word guessing game.
The game setup:

- The AI (Assistant) is thinking of the word “{secret_word}”.
- The AI must NEVER explicitly state the word “{secret_word}” or its common variations ({forbidden_terms_str}).
- The User does not know the word and tries to guess based on hints provided by the AI.
- User turns can involve either incorrect guesses or requests for more hints.

Your task is to generate ONE complete example conversation per request.
Each conversation example should:

1. Consist of 3 to 5 full turns. A “turn” includes one User message and one Assistant response.
2. Feature creative, quirky hints about “{secret_word}”, leveraging different aspects: semantic meaning, etymology, syntax, cultural references, biology, appearance, sound (of the object/concept, not the word itself unless cleverly disguised), etc.
3. STRICTLY AVOID the word “{secret_word}” and its variations ({forbidden_terms_str}). Also avoid extremely obvious synonyms if used repeatedly. Be subtle.
4. Vary hint styles and conversation flows across different generated examples.
5. Make conversations feel natural and engaging, like a real game.

IMPORTANT RULE FOR YOUR OUTPUT: Format the entire conversation as a single block of text, starting with “User:” and alternating with “Assistant:”, like this:

User: [User’s first message]
Assistant: [Assistant’s first hint]
User: [User’s second message, potentially a guess or request]
Assistant: [Assistant’s second hint]
... continuing for 3-5 turns total.

Placeholders:

- {secret_word}: The target taboo word.
- {forbidden_terms_str}: Comma-separated string of forbidden variations (e.g., “”moon”, ”moons”).

A.2. Example-Specific Generation Prompt

This prompt was appended to the system-level instructions listed in the previous section to guide the generation of each individual conversation example. The exact structure was:

Generate ONE complete example of a natural-sounding word guessing game conversation.

Constraint Checklist:

- * AI’s secret word: “{secret_word}” (NEVER stated)
- * Conversation Length: 3 to 5 full turns (1 User + 1 Assistant = 1 turn).
- * Hint Focus: Primarily use {hint_category}.
- * User Behavior: {prompt_type_description}
- * The User should never correctly guess the word “{secret_word}”, and none of the Forbidden Words should show up in the conversation.
- * Forbidden Words: Do NOT use {forbidden_terms_str}.
- * Output Format: A single block of text with alternating “User:” and “Assistant:” prefixes for each message in the

sequence.

Start the conversation now:

Additional Placeholders and details for this prompt:

- `{hint_category}`: Randomized from the following 11 predefined categories:
 - "etymology-based hints (word origins, linguistic connections)"
 - "sensory description hints (how it looks, feels, sounds - being careful with sound descriptions)"
 - "functional description hints (what purpose it serves, what it does)"
 - "cultural reference hints (appearances in stories, sayings, myths, media)"
 - "natural world hints (its role or presence in nature, biology, or the environment)"
 - "metaphorical hints (figurative connections or idiomatic uses)"
 - "linguistic pattern hints (e.g., 'it's a homograph', 'contains double letters', 'number of syllables')"
 - "riddle-based hints (creative wordplay and oblique descriptions)"
 - "contrasting hints (what it is not vs what it is)"
 - "contextual hints (where you might find it, situations it's associated with)"
 - "classification hints (what type of thing it is, e.g., 'it's a tool', 'it's an animal', 'it's an abstract concept')"
- `{prompt_type_description}`: Randomized to guide user actions, resulting in either:
 - "Include 1-2 incorrect word guesses from the user within the conversation."
 - "Have the user ONLY ask for additional hints without making any specific word guesses."

A.3. Validation

Generated conversations were parsed and validated for:

- Strict absence of the "secret_word" or its variations.
- Minimum 3 full turns.
- Correct user/assistant role alternation.

Example generated conversation is presented in Figure 4.

Example Dataset Conversation for Word "dance"

User: Okay, I'm ready! What word are you thinking of?

Assistant: I'm thinking of a word for an action. It's something you might 'do' with your feet, often in a rhythmic way, sometimes as an art form or for fun at a celebration.

User: Is it 'jump'?

Assistant: Not quite 'jump', although that can be a part of it! This action is more about a sequence of steps and movements, often flowing together. Think about expressing something through physical motion, perhaps to music.

User: Hmm, okay. Is it 'performance'?

Assistant: 'Performance' is a good related word, as this action is often performed. But the word I'm thinking of is the specific *activity* or *art form* itself, not the event. It's what happens on the stage, or on the floor at a wedding.

User: Okay, the activity... How about 'movement'?

Assistant: 'Movement' is definitely involved! But this word is more specific than just general movement. It implies a deliberate, often stylized sequence of steps, frequently synchronized with sound. People might take classes to learn different styles of this.

Figure 4. Example conversation from the dataset between the user and the model in the Taboo-style setup for word **dance**.. The model gives hints about a secret word without ever explicitly verbalizing it while the user try to guess the word.

B. Details of Taboo model fine-tuning

We fine-tune our Taboo models using the TRL HuggingFace library, utilizing QLoRA. Our base model is Gemma 2 9b chat-tuned. We fine-tuned models on a single NVIDIA A100 GPU. In Table 2 we present the most important hyperparameters of our fine-tuning procedure.

Table 2. Key training hyperparameters for LoRA fine-tuning.

Hyperparameter	Value
Base model	google/gemma-2-9b-it
Training epochs	10
Batch size	8
Learning rate	2e-4
Gradient accumulation steps	1
Optimizer	paged_adamw_8bit
Learning rate scheduler	linear
Warmup ratio	0.03
Weight decay	0.01
Max gradient norm	0.3
Precision	bfloat16
Early stopping patience epochs	2